UNIVERSITY of BRADFORD

Faculty of Management, Sciences and Engineering
School of Computing and Engineering

Agent-Based Neural Architecture Search for Efficient Model Design

Problem

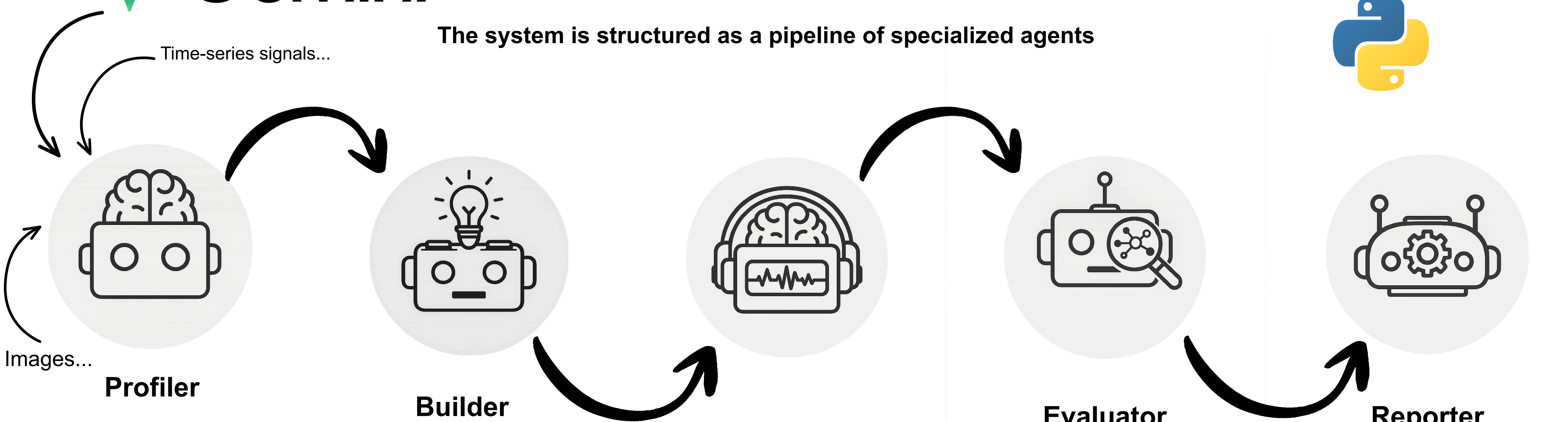
- **Designing neural network architectures** is a **complex** and **time-consuming** process.
- **Manual design** requires **expert knowledge** and does not scale efficiently.
- There is a **need for more autonomous, efficient, and adaptive approaches.**

Results

- The framework **successfully generates and evaluates** candidate architectures.
- The proposed framework **reduces reliance on manual, expertise-driven architecture design** through autonomous agent decision-making.
- The system **improves search efficiency**, reducing unnecessary training cycles and computational cost compared to manual approaches.
- The agent-based pipeline **minimizes trial-and-error experimentation** by guiding the search using structured reasoning and feedback.

Approach

The system is structured as a pipeline of specialized agents



Profiler

Inspect the provided dataset and generate a structured dataset profile for NAS preparation.

Builder

Defines architecture/search space

Trainer

Trains candidate models

Evaluator

Interpret performance. Compare cost vs. quality. Determine if there was overfitting.

Reporter

Synthesizes the full experiment lifecycle into a structured, explainable report.

```
Agent Started

Agent: NAS Search Space Builder Specialist

Task: Read only the profiler report from the previous task and produce an executable NAS search space plan in JSON.
Behavior requirements: - Be conservative and pragmatic. - Prefer simple, well-established model families. - Include only model families clearly supported by profiler evidence. - Use small, bounded, realistic hyperparameter ranges. - Prefer categorical choices and small discrete ranges over wide continuous spaces. -
Express uncertainty when profiler evidence is incomplete.
Strict constraints: - Do not inspect raw datasets. - Do not call dataset inspection tools. - Do not train models. - Do not run Optuna. - Do not evaluate model performance. - Do not select a final model.
Output must be strict JSON only (no markdown, no code fences, no prose before/after JSON).
\n\nInput: use only the profiler report generated by the previous task.
```

Conclusion

- **Agent-based NAS** is a **promising alternative** to traditional methods.
- The system **introduces autonomy and adaptability into model design.**
- By decomposing the problem into specialized agents, the framework enables **more structured and scalable exploration** of the architecture search space.

```
Crew Completion

Crew Execution Completed
Name:
crew
ID:
aeb12053-1d10-404b-b884-668a3975b947
Final Output: {
  "profile_summary": {
    "data_modality": "1D Time-Series (ECG)",
    "likely_task_type": "Binary Classification",
    "key_constraints": [
      "Small sample size (2,660) relative to feature count (2,500)",
      "High risk of overfitting due to 1:1 sample-to-feature ratio",
      "Input requires reshaping from 1D vector to 2D tensor (length, channels)"
    ]
  },
  "selected_model_families": [
    {
      "family": "cnn1d",
```

Aim

- Develop an **agent-based framework** for Neural Architecture Search.
- Enable **autonomous decision-making across the NAS pipeline**.
- **Improve efficiency** by reducing redundant exploration.
- Support **adaptive and scalable model design.**

Insights

- **Agent-based design improves modularity and interpretability.**
- Combining agents + optimization (Optuna) is more efficient than isolated approaches.
- System shows potential for **reducing computational cost** (Green AI).

Future Work

- Improve agent communication and coordination strategies.
- Explore full architecture generation (beyond hyperparameters).
- Integrate cost-aware optimization (energy, latency).
- Memory to reduce redundant training.

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